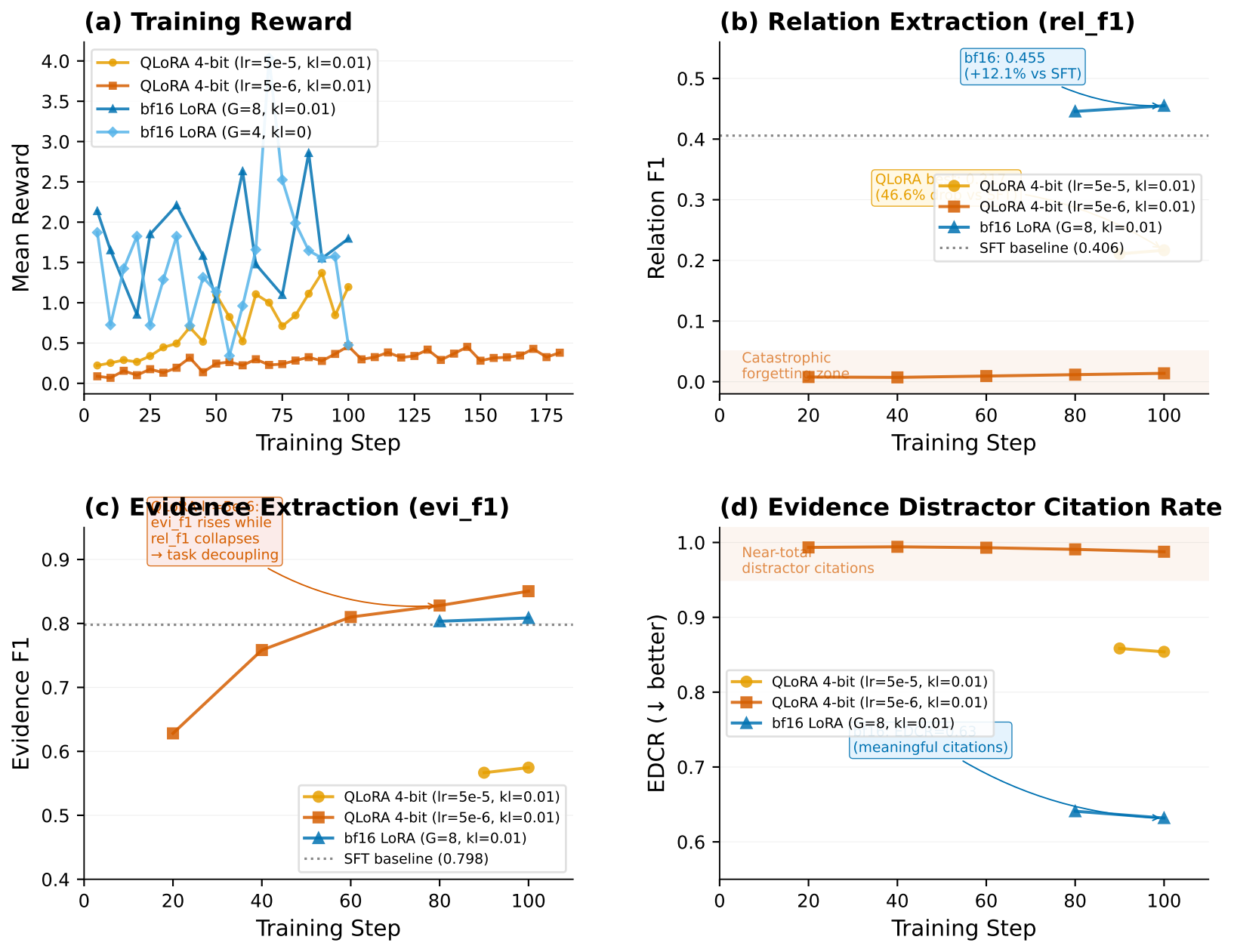
(a) Training Reward
Mean Reward
Training Step
QLoRA 4-bit (lr=5e-5, kl=0.01)
QLoRA 4-bit (lr=5e-6, kl=0.01)
bf16 LoRA (G=8, kl=0.01)
bf16 LoRA (G=4, kl=0)
(b) Relation Extraction (rel_f1)
Relation F1
Training Step
bf16: 0.455
(+12.1% vs SFT)
QLoRA 4-bit (lr=5e-5, kl=0.01)
QLoRA 4-bit (lr=5e-6, kl=0.01)
bf16 LoRA (G=8, kl=0.01)
SFT baseline (0.406)
Catastrophic
forgetting zone
(c) Evidence Extraction (evi_f1)
Evidence F1
Training Step
evi_f1 rises while
rel_f1 collapses
→ task decoupling
QLoRA 4-bit (lr=5e-5, kl=0.01)
QLoRA 4-bit (lr=5e-6, kl=0.01)
bf16 LoRA (G=8, kl=0.01)
SFT baseline (0.798)
(d) Evidence Distractor Citation Rate
EDCR (↓ better)
Training Step
Near-total
distractor citations
QLoRA 4-bit (lr=5e-5, kl=0.01)
QLoRA 4-bit (lr=5e-6, kl=0.01)
bf16 LoRA (G=8, kl=0.01)
(meaningful citations)